The Case for Establishing a Collective Perspective to Address the Harms of Platform Personalization

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ABSTRACT

Personalization on digital platforms drives a broad range of harms, including misinformation, manipulation, social polarization, subversion of autonomy, and discrimination. In recent years, policymakers, civil society advocates, and researchers have proposed a wide range of interventions to address these challenges. In this article, we argue that the emerging toolkit reflects an individualistic view of both personal data and data-driven harms that will likely be inadequate to address growing harms in the global data ecosystem. We maintain that interventions must be grounded in an understanding of the fundamentally collective nature of data, wherein platforms leverage complex patterns of behaviors and characteristics observed across a large population to draw inferences and make predictions about individuals.

Using the lens of the collective nature of data, we evaluate various approaches to addressing personalization-driven harms currently under consideration. This lens also allows us to frame concrete guidance for future legislation in this space and advocate meaningful transparency that goes far beyond current proposals. We offer a roadmap for what meaningful transparency must constitute: a collective perspective providing a third party with ongoing insight into the information gathered and observed about individuals and how it correlates with any personalized content they receive – across a large, representative population. These insights would enable the third party to understand, identify, quantify, and address cases of personalization-driven harms. We discuss how such transparency can be achieved without sacrificing privacy and provide guidelines for legislation to support the development of this proposal.

CCS CONCEPTS

• **Applied computing** → **Law**; • **Information systems** → **Personalization**; **Online advertising**.

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KEYWORDS

discrimination, polarization, filter bubble, echo chamber, algorithmic decision making

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1 INTRODUCTION

Platforms' ability to personalize content presented to their users has recently given rise to high-profile revelations and controversies—including the Facebook-Cambridge Analytica data scandal [33]; the emotional contagion experiment to influence Facebook users' moods [75]; research findings that leading platforms discriminate in their presentation of job and housing ads on the basis of race, sex, and age [68, 103]; and, most recently, the Wall Street Journal's investigative reporting on The Facebook Files [65].[1] Following the congressional testimony of Facebook whistleblower Frances Haugen in October 2021, a rare bipartisan response signified that "Facebook and Big Tech are facing a Big Tobacco moment" [70], with voices in the United States and around the world calling for stronger regulation of platforms [105].

Economic, social, and cultural activities are increasingly mediated by platforms [100], representing a shift "from industrial to information capitalism" [121]. To manage and leverage the growing amount of personal data they possess, platforms have developed and implemented artificial intelligence and machine learning algorithms, which, in turn, demand large volumes of data as inputs [6, 18]. Platforms harness their ability to capture, analyze, and act upon data regarding the behavior of large groups in a wide range of different ways. Leveraging personal data collection and analysis techniques, they are able to detect patterns of behavior and previously unanticipated clusters of users, make predictions about how individuals and groups of individuals will respond to personalized content, infer deeply personal attributes that an individual has not expressly disclosed, and act upon these predictions and inferences to influence behavior [50]. Such personalization – i.e., the ability to show each user content specifically chosen for

[1] In this article we use the term *platform* to refer to entities that collect, store, process, analyze, or act upon data pertaining to individuals (for example, in the provision of content, services, recommendations, or ads), and whose presence is primarily in the digital realm [29, 71, 99, 100]. We use the term *users* to denote individuals who use the services of the platforms. The term *individuals* describes people (who have not necessarily signed up to use a certain platform or agreed to its terms of service). Finally, the term *data ecosystem* refers to platforms, individuals, and any other entities participating in exchanging, transacting, and acting on data pertaining to individuals [29, 121].

them – can benefit users, but it also contributes to a broad range of data-based harms, including misinformation [106], manipulation [118], social polarization [12], subversion of autonomy [114], and discrimination [8, 45, 117, 119]. In response to these challenges, policymakers, civil society advocates, and researchers around the world have increasingly turned their attention to issues of data privacy and platform governance [21, 22, 31, 122].

In this article, we argue that the emerging toolkit reflects an individualistic view of personal data and data-driven harms and that such a framing will likely fail to adequately address the harms stemming from platform personalization. We assert that, instead, interventions must be grounded in an understanding of the fundamentally collective nature of data – i.e., the recognition that the personalized content that one person receives is strongly driven by rich data gathered about others around the globe [124]. Many personalization-driven challenges such as social polarization and discrimination cannot be defined with respect to one isolated individual; these harms, as well as the ability to define and detect them, inherently exist within a broader social context [78]. They also cannot be understood and addressed when the only parties currently with the capability to possess a comprehensive view of personalization processes are the platforms themselves.

Without meaningful, effective transparency, society lacks the essential tools to properly understand the role that personalization plays in generating and amplifying various harms. For instance, there remains considerable uncertainty regarding even basic questions, such as whether personalization is contributing to polarization or defusing it [19]. Critically, adequate transparency requires a far more expansive intervention than what has been contemplated in recent legislative proposals – e.g., disclosing ad targeting criteria or sponsorship details as in the Honest Ads Act [63], creating databases of ads divorced from the personal information of those who received them as in the Digital Services Act [42], or focusing primarily on ads (and not other personalized content) as in the Social Media DATA Act [108]. We offer design goals for achieving meaningful transparency: providing a third party with ongoing insight into the information gathered and observed about individuals and how it correlates with any personalized content they receive, across a large, representative population. We discuss how such transparency can be achieved without sacrificing privacy and provide guidelines for new legislation to support the development of such transparency.

The article proceeds as follows. Section 2 describes the structure and incentives of the data ecosystem, introduces new terminology to describe the flows of content between users and platforms, and highlights ways in which data about individuals is inherently collective. Section 3 uses this lens of the collective nature of data to evaluate various regulatory and technical approaches to address personalization-driven harms. Section 4 calls for new legislation to create effective transparency mechanisms and offers principles for their design, with the goal of generating a collective perspective that would enable a third party to analyze the effects of personalization across a large, representative population of users while protecting individual privacy.

2 THE STRUCTURE OF THE DATA ECOSYSTEM

In this section, we provide an overview of the structure of the data ecosystem and the incentives driving platforms' behavior, introduce terminology describing the flows of content between users and platforms and the feedback loop they create, and demonstrate why it is critical to recognize the collective nature of data when considering the suitability of interventions to address personalization-driven harms.

2.1 The Outgoing and Incoming Vectors

Platforms have powerful incentives and capabilities to design services, content, and interfaces to increase opportunities for widespread data collection and analysis and to use this information to present individuals with personalized content [28]. These activities are enabled by information flows in two directions. Along what we call the *outgoing vector*, platforms collect vast quantities of data about users and their activities [110], including interactions users have directly with the platform, interactions between users, and users' online activity beyond the platform. For example, when a user signs into a third-party service using their Facebook credentials, Facebook is made aware of their activity [112], and, when a Facebook user visits a web site that embeds a Facebook 'like' button, Facebook collects information about that visit regardless of whether the user clicked the 'like' button [83]. In some cases, platforms collect information about users' offline activity including information about their shopping habits, credit scores, public records such as census data and voter records, and more [4, 60, 77, 120]. With this information, platforms compile detailed user profiles, and, in some cases, they enable access to third parties as well [34, 93].

In the other direction, information flows from platforms to users, in what we call the *incoming vector*. This encompasses all content that platforms present to users based on the detailed profile the platform has created about them [123], such as suggesting groups to join, presenting posts or videos to users in their feed (including the order in which they are presented), and providing suggestions for other content they may be interested in (e.g., news articles, physical gatherings, and other users to connect with). Such personalization aims to increase user engagement and time spent on the platform – thereby enabling the platform to collect more data and to present more accurately tailored personalized content.

As illustrated in Figure 1, the outgoing and incoming vectors create a feedback loop: data collected along the outgoing vector is analyzed as a basis for personalization along the incoming vector, and individuals' interactions with personalized content from the incoming vector then generate more outgoing-vector data for the platforms to collect and analyze.

2.2 The Collective Nature of Data

Each platform user is associated with an extensive record of behavior, such as searches conducted, links clicked, posts liked, messages sent, photos posted, social connections formed, and more [73]. Although there is a tendency to think of such data as belonging to a single user [98], the reality is blurrier [14], as, for example, a message pertains to both sender and recipient, and a photo pertains

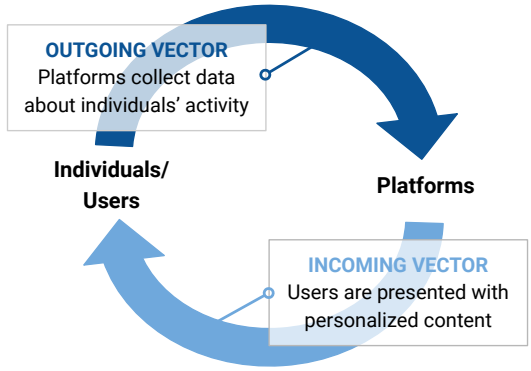

Figure 1: Schematic of the flows of information via the outgoing and incoming vectors.

both to poster and subject. Similarly, a Google search for information about a rare genetic disease may have implications not only for the searcher, but also for the searcher's genetic relatives [36].

In other words, platforms analyze data not to learn about users individually but to examine how they fit into patterns, clusters, and trends. As Salomé Viljoen argues, the relationships between users who belong to a group enable platforms to use data about one user to infer characteristics of another member of the same group, and it is often sufficient that a small minority has provided their data [121]. Solon Barocas and Karen Levy call these privacy dependencies and delineate three categories of dependencies by which the attributes of one individual can be inferred based on their social, physical, or electronic relationships with others: tie-based, similarity-based, and difference-based dependencies [14].

The most intuitive way by which one person's information provides details about another is where the second user is captured in the first user's data, in what Barocas and Levy term tie-based dependencies [14]. For instance, when Alice uploads a photo from a party she attended, the platform learns about her friend Bob who appears in the photo. The platform may also directly prompt Alice to provide information about Bob; for example, when downloading Facebook's Messenger app, users (sometimes unwittingly) give Facebook permission to access their entire contact list [24]. If enough of Bob's friends join the service, Facebook will be able to construct a web of Bob's social ties even though he himself has provided no information to the platform and may even prefer to avoid the platform altogether.

One individual's disclosure of information may enable the platform to learn something about another indirectly, whether by highlighting a similarity between the two individuals (in a similarity-based dependency) or by shedding light on the way that one user differs from others (in a difference-based dependency) [14]. By analyzing the behavior of an individual and comparing it to patterns of behavior across many users, platforms are able to make predictions about individuals and infer a broad range of personal attributes that they have not expressly provided [15, 124]. For example, when Alice shops on a platform, the platform gains knowledge of her personal attributes as well as her shopping habits. If another user with attributes similar to Alice's begins shopping on the platform, the platform may recommend to her some of the same products that Alice purchased. Similarly, if a new user has similar shopping patterns to Alice, the platform may infer that she shares (some of) Alice's personal attributes.

As another example, the detection of patterns across groups powers gender classification systems employed by platforms [74, 79]. These systems analyze user data such as pictures, videos, likes, and language patterns – drawing insights from the patterns among users who provided their gender in order to infer the gender of users who did not. More broadly, by analyzing users' behavior, interests, and social connections, platforms are often able to infer a broad range of personal attributes such as race, sexual orientation, income, political interests, and opinions [5]. Effectively, this means that no single individual can decide to withhold her data from platforms because an individual often cannot prevent a platform from learning about her by refusing to disclose certain data.

The collective, interdependent nature of personal data means that no single individual can decide on their own how much data they want to disclose to platforms, or what data they want to keep private. Yet, despite the substantial effects of inferences based on the relationships between users and the significant role that these relationships have played in the development of the data ecosystem, users' interests as a collective are currently severely underrepresented in regulatory discourse [121]. We contend that, to be effective, interventions in the data ecosystem must be grounded in a deep understanding of the collective nature of data.

3 APPROACHES TO OVERCOMING HARMS FROM INCOMING-VECTOR PERSONALIZATION

Recent sessions of Congress have produced an abundance of bills that aim to address the harms associated with platform personalization (e.g., [2, 25, 35, 49, 57, 63, 69, 94, 97, 107, 108]), alongside a vast array of other regulatory and technical proposals being introduced around the world. We analyze the tools employed within a selection of recent proposals as a reflection more broadly of emerging approaches to addressing personalization-driven harms.[2]

3.1 Liability and Enforcement Mechanisms

Enforcement of existing laws, such as anti-discrimination, consumer protection, and privacy and data protection laws, as well as reforms to existing liability protections for platforms, may address certain aspects of harmful personalization.

3.1.1 Anti-discrimination law. U.S. anti-discrimination laws, for instance, prohibit discrimination in ads for housing [45] and job opportunities [1, 117] based on protected attributes such as race, sex, age, religion, and more. In some cases, the content of the ads may not be inherently discriminatory, but the targeting criteria produce discriminatory effects by excluding certain groups on the basis of protected characteristics.

Pauline Kim and Sharion Scott identify at least three potential ways in which employment recruiting via targeted ad placement on

[2] While many of the harms these proposals seek to address are personalization-driven, others stem from non-personalization-related design choices made by platforms. For example, personalization of addictive design features or dark patterns can substantially amplify the harms they create, but the predominantly non-personalization-driven aspects of these features are not the focus of this article.

platforms can produce discriminatory effects [73]. The first occurs when advertisers select protected attributes as their targeting criteria. In a 2016 investigative report, ProPublica demonstrated that the Facebook ad targeting mechanism allowed the placement of housing ads that explicitly excluded users with African-American, Asian-American or Hispanic affinity from their targeting criteria [9]. The second involves the selection of targeting criteria based on seemingly mundane attributes, such as ZIP code or expressed interests, that are strongly correlated with, and in effect serve as a proxy for, one or more protected attributes [73]. While such targeting may result in discriminatory effects, it may be challenging to anticipate ex-ante [73]. The third way is through the use of a tool like Facebook's "lookalike audience" feature, which can identify a relevant audience based on a sample group, such as the employer's current workforce [85]. If the sample group is biased, this tool will likely produce an audience that reflects the same bias [110].

A strong argument can be made that targeting of the first type is illegal [1, 117]; however, further proof is likely to be needed to determine whether targeting of the second or third type was unlawfully discriminatory [73]. Further, detection of unlawful instances of ad targeting is limited by a lack of transparency into platforms' incoming and outgoing-vector flows of information, which is necessary for investigating and obtaining evidence of discriminatory effects.

3.1.2 Consumer protection law. The U.S. Federal Trade Commission (FTC), exercising its authority to bring enforcement actions against companies that engage in unfair and deceptive trade practices and to launch investigations [48], has also been active in addressing incoming-vector harms [47]. For example, in December 2020, the FTC launched an investigation, issuing orders to nine social media platforms to provide information about how they determine which content is shown to which consumers, and the effects of their practices on children and teenagers, among other questions [46]. In addition, FTC staffers are reportedly exploring whether Facebook engaged in deceptive or unfair trade practices in light of whistleblower Frances Haugen's statements in September 2021 regarding the company's internal research showing knowledge of harms resulting from its personalization algorithms [84].

3.1.3 Privacy and data protection law. The enforcement of existing privacy and data protection regulations, such as the GDPR [101], may also address some personalization-driven harms. Personalization may be seen to encroach on privacy rights, for example by undermining individuals' right to be left alone [109, 125]. Data protection principles, such as data minimization and purpose limitation (e.g., Articles 5(1)(b) and 5(1)(c) of the GDPR), likely serve to curb platforms' ingestion, use, and retention of large quantities of fine-grained user data in order to target highly personalized content. Scholars have observed that these principles are incompatible with the big data analytics at the heart of platform personalization because they require platforms to inform their users of the specific nature of future processing of data which, when using personalization algorithms, may be inherently unforeseeable at the time consent is given [64, 128]. In addition, data controllers and processors must demonstrate an applicable legal basis to justify the processing of personal data, such as consent or legitimate interests (Article 6(1)(a) of the GDPR), for platform targeting, which may pose challenges particularly in contexts in which profiling and tracking persist across multiple platforms [43].

Also in tension with platform personalization is the right not to be subjected to fully automated decision-making processes that substantially impact individuals, unless one of several exceptions, such as explicit consent, is met (Articles 22(1)-(2) of the GDPR). With respect to such processing, the GDPR provides that an individual has the rights to "obtain human intervention" (Article 22), "express his or her point of view" (Article 13(2)(f)), "contest the decision" (Article 14(2)(g)), to know of their existence (Article 15(1)(h)), and to receive "meaningful information about the logic involved, as well as the significance and the envisaged consequences of such processing for the data subject" (Article 15(1)(h)).

3.1.4 Reforms to platform liability protection. For decades, platforms have enjoyed legal protection from liability for harmful content posted by users, under laws such as Section 230 of the Communications Decency Act [90] in the United States and Articles 12-15 of the e-Commerce Directive [37] in the European Union.

Growing calls to restrict or withdraw the broad protection of Section 230 have inspired bills such as the Protecting Americans from Dangerous Algorithms Act [97], the Justice Against Malicious Algorithms Act of 2021 [69], and the Health Misinformation Act of 2021 [57]. These proposals would amend Section 230 to remove platforms' liability protection in certain circumstances where they take an active role in promoting harmful content. For example, the Justice Against Malicious Algorithms Act of 2021 [69] would hold platforms accountable for recommendations they made knowingly or recklessly and which contributed to physical or severe emotional injury.

A proposal in the EU, set forth in the incoming Digital Services Act (DSA) [42], would require large online platforms to set up notice-and-action mechanisms enabling users to report content they believe is illegal (Article 14 of the DSA).

3.2 Individual Control via Notice and Consent

Many proposals focus on individual control-based interventions [30], including notice and consent [13]. Such approaches are likely to burden individuals with uninterpretable, empty choices, rendering the sense of control they convey a mirage [56]. An extensive body of behavioral research calls into question the effectiveness of notice and consent mechanisms [11, 16, 17, 62, 81, 102]. Research demonstrates that individuals often fail to read or understand the implications of platforms' terms of service [13]. Moreover, such agreements are contracts of adhesion precluding the ability of individual users to negotiate changes to their terms [54]. Individuals, as participants in a knowledge-based economy, often lack a meaningful choice to opt out of the use of digital platforms altogether. We also argue that such approaches fail to recognize data's collective nature and its role in the personalization process [121].

For example, control-based approaches are incorporated in the European Commission's DSA proposal [95]. Article 29 of the DSA requires very large platforms to provide notice in their terms of service that the content users are viewing has been algorithmically generated and to detail the main parameters used by recommender systems [44]. They must also allow their users to influence the parameters used by recommender systems, including providing at

least one option to opt out of recommendations based on profiling (Article 29(1)).

In contrast to an omnibus legislative proposal like the DSA that seeks to address a wide range of incoming-vector harms, regulatory proposals in the United States tend to focus on combating specific categories of harms, such as those stemming from platform experimentation and the filter bubbles created by platforms' manipulative presentation of personalized content, as we discuss in turn.

3.2.1 Experimentation. The unique position of platforms within the data ecosystem enables them to experiment with the presentation of different types of content and observe how various categories of users respond. Platforms continuously run such experiments, aiming to refine their personalization algorithms, boost the impact of content presented to users, and make ongoing changes to their interface in order to generate increased engagement [75]. However, individuals may not be aware that platforms are experimenting on them, or that the content they are seeing is based on present or past experimentation. Notably, Facebook's mood manipulation experiment – studying whether users' emotional state could be influenced by the content they were shown on the platform – sparked widespread criticism from civil society, academics, and regulators alike [61].

In response, one proposal, the Deceptive Experiences To Online Users Reduction (DETOUR) Act, would require platforms that conduct psychological or behavioral experiments on their users to obtain users' informed consent and to periodically disclose to users as well as to the general public any experiments being conducted by the platform [35]. Because its scope is limited to a narrow subset of personalization in the context of psychological or behavioral experiments, this proposal arguably fails to address harms from other similar types of testing used by platforms to refine their personalization of content to users [23, 66, 67]. Further, because the bill relies solely on tools enabling each individual to exercise control with respect to the content they see, it fails to recognize the collective nature of data. For example, if Alice has opted out of platform experimentation, but Bob, a friend of Alice, has not, Alice might still see content Bob has interacted with as part of the experiment, due to the socially-driven nature of many platforms' recommendations.

3.2.2 Filter bubbles. Scholars, politicians, and the media have sounded the alarm regarding platforms' role in creating echo chambers [111], or filter bubbles, in which users have little or no exposure to opinions or even news reports that contradict their beliefs [12]. "Interactional polarization" and social fragmentation are vital concerns [127], as deliberation, persuasion, and compromise with opposing views – central to democratic functions – are precluded by the very nature of the personalized experience each user encounters on online platforms [111].

It is against this backdrop that the Filter Bubble Transparency Act (FBTA) was introduced [49]. The bill would require large platforms to disclose to users that the content they are presented with and the order in which it is presented is determined by an algorithm and based on user-specific data, or inferences based on user-specific data (Section 3(b)(1)(A)). Additionally, it would require platforms to enable users to opt out of the filter bubble and instead view an input-transparent version of the platform, i.e., a newsfeed that was not algorithmically personalized based on user-provided content (Section 3(b)(1)(A), (B)). This approach is similar to the DSA's requirement for platforms using recommender systems to notify their users and enable them to opt out of seeing content based on profiling (Article 29 of the DSA [42]).

One question is how an alternative view not based on user-provided content would be generated. For instance, would users of social networks still see content posted, liked, or shared by their social contacts, or content from groups they belong to or pages they have liked (say, in reverse chronological order)? If so, the assumption that any one individual could single-handedly remove herself from the content promoted within the filter bubble while remaining active on platforms reflects a lack of understanding of the collective nature of data. This is because an individual who opts out would still see a newsfeed laced with content that personalization algorithms promoted to her social media contacts.

Without an overhaul of the current approach to control-based mechanisms, it is unlikely such mechanisms will translate into greater protection of individual autonomy. Effective consent-based mechanisms must ensure individuals are able to make meaningful and consequential choices regarding authorized uses of their data, including permissible types of personalization. Additionally, they must be presented with more than one viable option to choose from, the consent process must not be overly burdensome, and they must be adequately informed about the ramifications of each choice. Inasmuch as platforms' incentives remain fixed, however, countering the harmful effects of platform personalization will require entrusting a third-party body with a collective perspective as outlined below in Section 4.

3.3 Transparency Mandates

A third category of interventions includes mandates that platforms disclose certain information regarding personalization to third parties. The need for statutorily-mandated transparency was underscored by recent attempts by platforms to block third parties from collecting information about outgoing- and incoming-vector content. In August 2021, Facebook shut down the accounts of three New York University researchers who were initially granted access to conduct a study regarding political ads on the platform [39], on the grounds that they had violated the platform's terms of service prohibiting the use of automated scraping tools and, in particular, that such scraping posed risks to individual privacy [26]. Facebook has responded to criticism about its lack of transparency by making certain data available to researchers; however, researchers have noted that access has been too limited to enable effective study of harms such as disinformation and manipulation and that federal legislation mandating platform data sharing is urgently needed [59].

Many proposals promoting transparency require the disclosure of the targeting criteria selected by advertisers and other considerations introduced by the platforms in the actual presentation of ads, which can potentially address harms especially with respect to discrimination and disinformation.

3.3.1 Discrimination. The Algorithmic Justice and Online Platform Transparency Act [2] would mandate that platforms retain a record containing data about their algorithmic processes and provide access to the FTC upon its request (Section 4(a)(2)(C)). The bill also

states that if an algorithm promotes ads for services such as housing, education, employment, insurance or credit, the platform must also assess whether the algorithm creates a disparate outcome based on a protected attribute [2] (Section 4(a)(2)(A)).

3.3.2 Disinformation. Disinformation campaigns have interfered in democratic elections, engendered mistrust in democratic institutions and in democracy itself [51], and incited violence [113, 115, 116]. Although disinformation is not an exclusively personalization-driven harm, the harmful effects of disinformation are substantially amplified by platforms' ability to present such content to users who are more susceptible to believing and acting upon it. The spread of disinformation online can also indirectly impact individuals who do not actively participate on digital platforms.

Following findings of Russian involvement in the 2016 U.S. presidential election [89], the Honest Ads Act [63] was introduced, requiring platforms to accompany political advertisements with a clear statement disclosing who is financing them. The law also mandates that platforms maintain a publicly accessible database regarding various details pertaining to political ads placed (or requested to be placed) on them, including, inter alia, "a description of the audience targeted by the advertisement" (Section 8(j)(2)(B)). This requirement seeks to establish a collective point of view regarding the ability to detect personalization. However, because it requires disclosure of only the targeting criteria (as collected along the outgoing vector) and not data about the actual presentation of the content (as presented along the incoming vector), it would not enable a third party to detect correlations between outgoing- and incoming-vector content.

Further, the bill's focus on political ads renders it unlikely to prevent manipulation of political processes, due to the influence other types of content have on elections. For example, the Mueller report found that much of the disinformation spread online in the period leading up to the 2016 US presidential campaign did not appear in the form of ads [89].

3.3.3 Researcher access. Some proposed transparency requirements involve the creation of databases for use by researchers. For example, the DSA proposal requires platforms to create a repository of ads presented on their interface including a copy of the ad itself, information about the targeting criteria used and aggregate information about the number of users actually presented with the ad (but not information about their personal attributes) [42]. The DSA further requires platforms to provide vetted researchers with information that would enable them to identify systemic risks created by platform activity such as dissemination of illegal content or intentional manipulation of platforms' services. Since the DSA does not require disclosure of key outgoing vector data about users who saw the ads, however, it falls short of enabling analyzing correlations between outgoing- and incoming-vector data.

The Social Media DATA Act [108] would mandate that platforms provide academic researchers and the FTC with access to all ads placed by advertisers – together with details about the targeting criteria and mechanism (of the advertiser and the platform) and details about the demographics of the audience actually presented with the ads. The Platform Accountability and Transparency Act (PATA) [94] would enable researchers to submit research proposals to the National Science Foundation, and, upon approval, the relevant platforms would be required to provide the data requested. Additionally, the proposal would enable the FTC to require ongoing transparency about certain data, even if no particular request has been made by researchers.

3.3.4 Privacy protections. In many cases, transparency mandates are paired with safeguards to protect individual privacy when sharing data with third parties. Indeed, privacy is a central concern when platforms are required to disclose user data and is often cited by platforms as a rationale for denying data requests from third parties [94]. PATA includes provisions requiring researchers to submit their research results to the FTC prior to publication in order to ensure final research products do not compromise privacy or other confidential business information [94].

The Social Media DATA Act envisions that the FTC would establish a working group "tasked with providing guidance on how independent research using social media data can be done in a way that protects academic researcher independence and consumer's rights to privacy," [91] including consideration of "[u]nder what circumstances privacy preserving techniques such as differential privacy and statistical noise could be used" [108]. Differential privacy is a mathematical technique for adding statistical noise to computations to mask the influence of any particular individuals' data on the outcome [38, 58, 126], and we agree that it is well suited to the computation of aggregate statistics such as would be needed in order to audit for problematic personalization, as discussed in more detail below in Section 4.

Requiring transparency regarding targeting criteria and information about the actual presentation of ads – along with enabling third parties to explore the correlations between incoming-vector personalization and outgoing-vector demographic characteristics – could create a broad, collective perspective that would likely enable the study and detection of potential cases of unfair treatment, illegal discrimination, or disinformation.

3.4 Self-Regulation

Many platforms have adopted policies regarding removing, blocking or restricting of content. For example, Facebook implemented a third-party fact-checking program to limit the spread of disinformation by identifying the source and reviewing the content of suspect posts [86]. In addition, Facebook established an oversight board entrusted with the authority to make binding decisions about content Facebook should remove from its platform [92].

In 2018, the European Commission signed a Code of Practice on Disinformation together with leading platforms, such as Facebook, Google, Twitter and Mozilla, and later joined by Microsoft, TikTok, and advertisers [40]. The Code recognizes the harm caused by platforms' amplification of disinformation, and seeks to strike a balance between individuals' freedom of expression on one hand and the harms created by disinformation on the other [40]. The first-year assessment of the Code found that it served as a basis for dialogue between stakeholders and provided transparency into platforms' disinformation policies, but suffered from a lack of access to platform data that prevented third parties from assessing platforms' adherence to the Code [41].

While platforms' self-regulatory efforts may be a complementary step in the right direction, particularly because they may encourage standards which are more aligned with current technological practices, such efforts involve very little external oversight [27] and run the risk of being "little more than a symbolic activity". [20] Creating a transparency-increasing mechanism, which enables third parties to observe platforms' behavior and take part in tracking their adherence to the standards created by self-regulation, can potentially mitigate some of these concerns, as described in more detail below in Section 4.

3.5 Technical Approaches

A number of recent projects seeking to address the harms of personalization take a control-based perspective, seeking to keep each individual's data in a location controlled by that person, and allowing software under their personal control to dictate whether outside platforms and apps can gain access to their data. While there is both a role and a need for better control of data, we find that such an individualistic perspective ignores the fundamentally collective nature of data and thus would not be able to meaningfully intervene to prevent incoming-vector harms.

Francis Fukuyama et al. have recently begun to explore a structural intervention they call "middleware," software that would enable users to choose the type of content they want to see, how it should be ranked, and the sources they trust to present them with such content [52, 53]. Such an intervention is intended to dilute the power that platforms currently have over public and political discourse, and it is technologically situated to minimize friction with the existing ecosystem [52]. However, it is not clear how individual preferences would interact with platform-driven content promotion or personalization in their model. Furthermore, as framed, the middleware proposal does not seek to provide insight into patterns of personalization or their impacts.

A handful of recent technical projects have explicitly aimed to make incoming-vector personalization more transparent. For example, a number of carefully constructed studies, combining incoming- and outgoing-vector data at a fixed point in time reveal instances of problematic discriminatory presentation of advertising content by platforms [3, 4, 10, 76]. The limitation of such studies is that, because there is no general infrastructure for collecting such data, they require considerable time and effort to implement, and they are capable of providing visibility into only one isolated issue at one point in time.

Another approach, the Mozilla Rally project [88], allows individual users of the Mozilla Firefox web browser to sign up, volunteer information about themselves (i.e., outgoing-vector content such as demographic characteristics or answers to surveys), elect to allow Mozilla to gather content related to their browsing (such as the URLs of the pages they browse, page content, and how much time they spend on each page), and opt-in to allow pre-approved research projects access to their relevant data [87]. This effort, if widely adopted, could potentially provide broad, meaningful transparency into platform personalization from a collective perspective, due to its access to (some) incoming and (some) outgoing vector data. One downside is the project's lack of formal privacy guarantees for the potentially quite sensitive data that it gathers. However, the high-level idea is an incredibly promising model.

In summary, we find that many current proposals adopt an individualistic approach. This finding is consistent with many scholars' observations that privacy and data protection have traditionally been conceptualized as individual rights [82], largely focused on individuals' ability to control the flow of their data through the data ecosystem. This framing of data, however, ignores the current reality in which the process of datafication creates unjust results on a social level [121], and the fact that it takes analyses of massive amounts of data of many individuals to personalize content in a way that subordinates and manipulates individuals as well as generating collective harms [121]. An individual acting on their own cannot counteract either end of this problem: they cannot, on their own, effectively withhold their data along the outgoing vector and they cannot effectively extricate themselves from the harms of incoming-vector personalization.

4 RECOMMENDED DESIGN PRINCIPLES FOR EFFECTIVE INCOMING-VECTOR INTERVENTIONS

In this Section, we propose a path forward for addressing the harms of personalization. In particular, we argue there is a need for a particular form of transparency that we refer to as a collective perspective: transparency that allows visibility into correlations between the incoming and outgoing vectors with respect to a large population of users.

4.1 What Information Is Needed to Achieve Meaningful, Effective Transparency?

Meaningful transparency must constitute visibility into the personalized content presented to a large number of individuals, not just one or a handful. Some of the harms induced by incoming-vector personalization are only definable within a broader social context. More crucially, though, given any definition of what constitutes problematic personalization, the data of only a single person or a small number of people cannot generally be used to determine the presence or extent of the problem. For example, if one wished to show that a particular ad for housing was being displayed in a manner that disproportionately excluded Black individuals, it would not be enough to observe that the ad was shown to a particular white person or was not shown to a particular Black person. Instead, one would need to know the rate of display on a representative sample of the relevant white and Black populations, and one would need enough observations such that measured differences in the rate of display would be statistically significant. Similarly, if one wished to detect content that was being promoted too rapidly or diverging to extremes, one would need to analyze a broad sample of individuals' incoming-vector content.

The precise number of people's perspectives needed in order to detect problematic personalization depends on the number of types of harms one wishes to audit for (i.e., when there are more questions to be studied, one must increase the number of observations in order to maintain statistical validity of the conclusions), the sizes of the populations one wishes to study (i.e., if one wishes

to detect discrimination against a tiny group, it may be difficult to get enough observations of that group), the prevalence of the problematic phenomenon (i.e., one needs many observations in order to see rare events), and the severity of the phenomenon one wishes to detect (i.e., it requires fewer observations to detect extreme discrimination than subtle discrimination). In practice, the actual number of individuals needed to form a useful collective perspective could range from the dozens to the tens or hundreds of thousands. Past experiments provide some insight into the size of cohorts that have been required to detect specific instances of problematic personalization; for example, one study used a cohort of tens of thousands to hundreds of thousands participants [4].

Additionally, meaningful transparency must expose patterns and correlations that relate outgoing-vector content (such as individual characteristics and actions taken) to incoming-vector content at an aggregate level. Visibility into only incoming-vector content would be blind to how content-presentation algorithms employed personalization. It is the individual characteristics and behaviors revealed along the outgoing vector – indicating individuals' personal attributes – that form the basis of such personalization. Hence, the ability to relate the outgoing vector to the incoming vector is a crucial component of meaningful transparency.

Insights into incoming-vector personalization must be ongoing, rather than a one-off measure, as personalization algorithms and their content (and thereby their harms) are constantly changing and evolving. Furthermore, some concerns, such as platforms' promotion of increasingly polarized content, have an inherent longitudinal aspect.

In sum, meaningful transparency requires far more than disclosing ad targeting or funding criteria, or creating a database of ads. To be effective, transparency must generate a collective perspective, with ongoing insight into the information provided and observed along the outgoing vector and how it correlates with any personalized content received along the incoming vector, across a large, representative population.

4.2 What Body Could Be Tasked with Establishing a Collective Perspective?

Currently, platforms are the only actors that hold a collective point of outgoing- and incoming-vector content of many users. Past analyses, such as by Lina Khan and David Pozen, suggest that platforms' incentives are so misaligned with those of individual users and the public at large that platforms should not and cannot be solely responsibile for detecting, measuring, and mitigating the harms they magnify [72]. It is therefore necessary to explore alternative bodies that could be entrusted with the collective perspective.

A body using a collective perspective to detect, measure, or respond to problematic personalization along the incoming vector would need to be trusted, in at least two senses. It would need to be trusted to carry out its duties of observation or intervention in the best interests of individuals and of society, and the analyses it might perform could involve quite sensitive information. Such a system should be designed with technological safeguards in place so that it is able to make the required measurements (such as quantifying gender disparity in the delivery of a certain type of ad) without any third party accessing the raw data. The Social Media DATA Act [108], for example, recognizes the potential of modern technology to resolve seeming conflicts between transparency and privacy [107].

Often transparency does not require direct access to individuals' sensitive information but just to statistical aggregates that can be computed with small, intentional perturbations in order to provide formal privacy guarantees. Local differential privacy is one tool that can be used to allow the computation of accurate aggregate statistics (such as the level of correlation between an ad being shown and the race of the viewer) on the basis of personal data to which a large but controlled amount of random noise has been added.

Secure multiparty computation tools additionally provide a modern cryptographic toolkit that can remove the need for a monolithic body to be entrusted with correctly and safely carrying out computations related to the presence of problematic personalization [80]. Instead, responsibility for carrying out the duties related to the collective perspective could be shared across a few trusted parties, and as long as most or even some of them were following the protocol, none would be able to corrupt the computation or gain inappropriate access to personal information.

Statutory and contractual safeguards could provide an additional layer of protection. There are a number of possible models for how the information needed to establish this collective perspective could be sourced by a third party, including potentially by directly intermediating between individuals and platforms, receiving information primarily from individuals, or receiving information primarily from platforms. The governance, funding, and structure of the entity could also take different forms, ranging from a government body to a highly-regulated for-profit or non-profit service. In addition, as we detail below, there are a number of possible choices of entities that might be granted access to the insights afforded by the collective perspective.

The proposed EU Data Governance Act (DGA) [96] provides one useful model for establishing trustworthy intermediating bodies. Under the proposal, data intermediaries would maintain neutrality, and would use the data only for promoting its lawful exchange. Their business model must "assure that there are no misaligned incentives that encourage individuals to make more data available for processing than what is in the individuals' own interest" [96]. Furthermore, they would owe a fiduciary duty to those data holders whose data-sharing they facilitate. The DGA would also recognize data cooperatives that would support users in their data-sharing and serve as a tool in advancing users' ability to make informed and meaningful choices over their data and its sharing, inter alia, by enabling "mechanisms to exchange views on data processing" that would best represent members' interests [96]. Such a body could potentially also be positioned to establish the needed collective perspective.

4.3 How Can Legislation Support the Establishment of the Necessary Collective Perspective?

Legislation must take an active role in establishing or identifying an intermediating body that can hold the collective perspective, ensuring that it has unencumbered access to the necessary data,

establishing mechanisms for the harms that it surfaces to come to light, and providing enforcement mechanisms against those harms. It should also tie the hands of the intermediating body, restricting its ability to share any data it is granted access to (and derivatives of such data), and mandating the use of modern cryptographic and statistical techniques to minimize the collection and exposure of sensitive data.

In line with the transparency mandates discussed in Section 3.3, legislative intervention will also likely be necessary in order to oblige platforms to cooperate with the monitoring and data collection required in order to establish the collective perspective. In this respect, legislation must ensure the intermediating body's access to the information it needs. For example, one might consider laws allowing users to install software that enables a third party to collect information about their interaction with a platform. Currently, platforms restrict users' ability to share content outside the platform in their terms of service and do not allow third parties to scrape content from the platform [104]. Care must be taken to ensure that privacy and security concerns (whether real or fictional) and that corresponding laws, such as the Computer Fraud and Abuse Act [32], will not be used as an excuse to hamper the effectiveness of the intermediating body.

Legislation would also determine who would have the right to query or access the collective perspective. Under one model, in analogy to the approach taken by PATA, it could be made accessible for interrogation by academic researchers who are subject to oversight by institutional review boards and have applied for and been granted approval to carry out studies on the data. Researchers who detect harmful personalization could share their research findings with the appropriate oversight body in support of investigatory and enforcement actions. Alternatively, or additionally, access to the collective perspective could be made available to journalists for investigative reporting purposes. An advantage of these models is that granting academic researchers and journalists access to the collective perspective could enable identification of newly-emerging informational harms, in addition to violations of existing law. A third model would make the collective perspective available to a government agency with investigation and enforcement authority, such as the US Federal Trade Commission, the US Department of Housing and Urban Development, or the US Equal Employment Opportunity Commission (EEOC). In cases where such a body uncovered illegal personalization it could file a complaint, as the EEOC did alleging Facebook facilitated the discriminatory presentation of job ads [7]. Finally, an independent, cooperative entity, such as a data cooperative or data trust, could be established with the explicit purpose of monitoring for impermissible personalization.

4.4 What Is the Expected Impact of the Collective Perspective?

The collective perspective, once established, would shed light on the mechanisms by which personalization contributes to known harms, enable quantification of the severity of harms, and potentially identify previously unrecognized harms. This would provide a basis for deliberation among academics, policymakers, and society at large, grounding in evidence the debate around pressing questions such as: how severe is discrimination in digital advertising of housing opportunities, and what role does platform personalization play? Does personalization on the basis of inferred characteristics contribute more to the amplification of misinformation than personalization on the basis of characteristics explicitly provided? How significant is the contribution of personalization to the rapid spread of incendiary content?

Once a collective perspective is established, legislation could prescribe enforcement against problematic personalization, such as criminal or civil penalties for platforms, or flagging, deprioritizing, or blocking of harmful content. Adherence to norms regarding personalization could also be promoted by providing key results of telemetry to individual users, regulators, or the public. These norms could be reinforced by legislation but could also be adopted in community norms by groups of users who wish to adhere to certain standards that go beyond what is required by law. For example, a group of users may prefer not to see content that has been personalized based on their political affiliation or that employs gender-stereotyped personalization. Meaningful transparency into personalization could give people the power to pressure platforms to offer such options.

In summary, we call for legislation to establish a collective perspective that will enable society to detect, study, quantify, and respond to personalization-based harms. Without such intervention, problematic personalization will likely continue to cause harm unchecked, and even largely unobserved.

5 CONCLUSION

Evaluating the emerging regulatory and technological toolkit that has been proposed to address the harms stemming from platform personalization, we argue that most of the current proposals will likely be ineffective due to their failure to recognize the collective nature of data. We call for an intervention based on a new approach to transparency, which we refer to as a collective perspective. This approach would afford third parties with continuous visibility into correlations between the outgoing vector (data flowing from a user to a platform) and the incoming vector (content presented to a user by a platform), across a large, representative population of users. We outline how legislation can support the establishment of a collective perspective while providing strong safeguards for individual privacy.

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